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Predicting daily stock price directions with deep learning models

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ABSTRACT

In this paper, deep learning models like LSTM, CNN and RNN were explored to predict the direction of daily stock price changes. Since stocks in the same industry sector are highly correlated, we propose to replace individual stock with their corresponding Exchange Traded Sector Funds (ETFs) and S&P 500. We show that this replacement of individual stocks by the corresponding ETF can be justified due to high degree of returns correlation and small Hamming distances of trading signals across both input and output for stocks in the same sector. We considered historical daily data spanning 25 years (2000 to 2024) on major sector ETFs, some of their components and the S&P-500 index. Our results show that LSTM consistently outperformed CNN and RNN and traditional machine learning models. As a trading strategy, LSTM significantly out-performed buy-and-hold strategy for stocks in the Technology and Consumer Durables sectors, but offered very minor improvement for stocks in other sectors.

Our results suggest that predicting daily stock price directions with deep learning models should not be used for most stocks unless these stocks are in Technology or Consumer Durables sectors and LSTM-based models are used.

1. Introduction

Predicting the direction of stock price movements has been the most difficult yet essential task for investors, financial analysts, and researchers. The market is inherently volatile and, therefore, relies on the conventional methods of forecasting with statistical models and technical analysis. The inherent volatility of markets has traditionally prompted reliance on conventional forecasting methods like regression analysis (Siew & Nordin, 2012), exponential smoothing (de Faria et al., 2009), auto-regressive integrated moving average (ARIMA) (Ariyo et al., 2014), and technical indicators. These methods often fail to capture the complex dependencies and patterns within financial time series data. However, deep learning models have started gaining significant attention for being able to learn detailed patterns and dependencies over extended periods of time, making them a promising approach for stock price prediction (Cao, 2024; Kundu & S, 2024). For a comprehensive survey on binary neural network, see Qin and Gong (2020).

This paper explores the application of deep learning techniques like Long Short Term Memory (LSTM), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) to forecast daily stock movements of Exchange Traded Funds (ETFs) and the S&P 500 stocks for the "24 hours" (close-to-close) trading session.

LSTM is a type of RNN used for sequential data with memory cells to retain important information (Sherstinsky, 2020; Yu et al., 2019), making it effective for long-term trend prediction, especially for tasks like financial time-series prediction (Yadav et al., 2020). CNN is mostly used for image processing but also useful in detecting short-term patterns in financial data using filters(Hoseinzade & Haratizadeh, 2019). RNN, on the other hand, is a neural network for sequential data where each output depends on previous inputs. However RNN has issues retaining long-term memory (Wang et al., 2024).

These network architecture models are efficient in pattern recognition of the time series data. LSTM even though an old method, is still a profitable trading strategy and is useful in financial forecasting. It has proved the same for the stock market in China as well (Chen et al., 2015).

This paper addresses the following questions:

1. **Can sector ETFs accurately represent the stock movement directions of its constituent stocks?**
2. **How effectively can deep learning models predict the direction or trends of ETF sectors stock prices and help design trading algorithms?**

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3. **How do Deep Learning models for ETF sectors compare with traditional models like Logistic Regression and Random Forest?**
4. **Which neural network model is the best to predict stock movement directions?**
5. **Is it beneficial to combine these neural models as ensemble models?**
6. **Which sectors (by extension sector stocks) can be predicted by Deep Learning architecture models?**
7. **How effective is parameter tuning in predicting accuracy and in designing a better trading strategy?**

We suggest to address this as follows. Stocks do not exist in isolation and usually reflect some industry. Therefore, just like in clustering, we group stocks into clusters based on some similarity metric and replace the analysis of individual stocks by analyzing a typical "centroid" stock. A natural way to cluster is to note that most stocks belong to some industry and stocks in the same industry tend to be highly correlated. It would seem plausible to assume that deep learning models would predict the same sequences of daily price change movements for all stocks in the same sector.

Therefore, we suggest to proceed as follows. We will confine our attention to large-cap S&P 500 stocks. There are 9 major industry sectors that comprise these stocks, such as energy, financials, healthcare, etc. For each of these sectors, we have the exchange-traded funds (ETFs).

ETFs (exchange-traded funds) are securities that trade like stocks on exchanges, typically tracking the performance of a specific index, sector, or asset class. ETFs are particularly interesting for deep learning applications because they represent groups of stocks rather than individual companies. This makes ETFs more stable and potentially useful as reference points (or "centroids") for predicting overall market movements which is elaborated in our paper as well. Several researchers (Chun et al., 2020; Solıman et al., 2023) suggested that ETFs could act as indicators for sector-wide trends, helping investors understand broader market movements. To improve efficiency and interpretability of deep learning models in finance, active feature acquisition can be useful as well (Guney et al., 2025). In his study of ETFs,Zhang et al. (2020) proposed a clustering-based asset allocation approach using ITML, integrating financial insights with mathematical metrics to construct diversified ETF portfolios that outperform random selection in risk-adjusted returns.

In this study, the focus is on the following sector ETFs which represent various markets along with some of their corresponding component stocks:

1. **XLB** - Materials (EcoLab, Sherwin-Williams, Air Prod. Chemicals)
2. **XLE** - Energy (ExxonMobil, Chevron, Schlumberger)
3. **XLF** - Financials (J.P. Morgan and Chase, Bank of America, Citi)
4. **XLI** - Industrials (Honeywell, Union Pacific, Raytheon)
5. **XLK** - Technology (Apple, MSFT, NVIDIA)
6. **XLP** - Consumer Staples (Procter & Gamble, Coca-Cola, Pepsico)
7. **XLU** - Utilities (NextEra Energy, Duke Energy, Southern Co.
8. **XLV** - Healthcare (UnitedHealth Group, Johnson & Johnson, Merck)
9. **XLY** - Consumer Discretionary (Amazon, Home Depot, Starbucks)
10. **SPY** - S&P 500 Index (Intel, Walmart, Cisco)

The first 9 ETFs are sector ETFs. The last ETF "SPY" is the ETF for the S&P-500 market-capitalization index representing 500 of the largest publicly traded companies in the United States. The SPY ticker is assigned to reflect the S&P 500 index, serving as a proxy for the broader market (Vijh, 1994).

Our focus in this study was particularly on large cap stocks as it ensures data reliability, trading feasibility, and relevance for ETF-based modeling and investment strategies. Extending this approach to midcap stocks, and especially to small cap stocks may require addressing additional challenges, such as liquidity, higher volatility, and possibly weaker correlations with sector-level proxies.

As a trading strategy, we use the passive buy-and-hold strategy where an investor buys a stock holds this stock for an extended period, regardless of market fluctuations. But certain researchers (Beniwal et al., 2023; Hui & Chan, 2014) have shown that use of some strategies can be effective in beating Buy-and-Hold method which also aligns with our results.

In this paper, we show that when applied to ETFs, LSTM performed much better than other deep learning methods and better than Buy and Hold. We believe that this provides a very strong evidence that LSTM provide an effective way to predict daily price movement direction for a typical stock with a profitable trading strategy. Our trading strategy acts on the daily directional signal from the deep learning model. If the signal is positive ('+'), a long position is entered or maintained for that asset. If the signal is negative ('–'), any open position is closed, and the portfolio moves to cash. No short-selling or leverage is used. Typical transaction costs of 0.05–0.1% per trade (Quantified Strategies, 2025) are applied to all results. All cash balances were assumed to remain idle and did not accrue any interest during the evaluation period.

This paper is organized as follows. Section 1 introduces the study, outlining the challenges of stock price prediction and motivating the use of deep learning models such as LSTM, CNN, and RNN. Section 2 describes the proposed methodology, including data selection, preprocessing, feature engineering, and the implementation details of the deep learning architectures. Section 3 presents the results and analysis, reporting the performance of the models and comparing them with a Buy-and-Hold benchmark. Section 4 discusses the findings, providing interpretation of the results together with additional performance metrics and visualizations. Section 5 highlights the study's contributions and limitations and situates the work within the context of prior research to demonstrate its novelty. Finally, Section 6 concludes the paper by summarizing the key findings and emphasizing the potential of LSTM for stock market prediction. Supplementary details, extended performance metrics, and additional visualizations are provided in Appendix A sections.

2. Methodology

The application of deep learning models in stock market prediction has been rapidly evolving. In recent years, deep learning models have emerged as powerful tools for financial forecasting due to their ability to learn intricate patterns and temporal dependencies (Heaton et al., 2018). The existing baseline models that align with the latest advancements in the industry are explained in papers as well (Jiang, 2021). Mehtab and Sen (2020) develops and evaluates deep learning models, including CNN and LSTM variants,demonstrating that a convolutional LSTM model with shorter input data achieves the highest accuracy. LSTMs, in particular, have proven effective in capturing complex relationships in financial data, making them suitable for tasks like stock forecasting (Althelaya et al., 2018), portfolio optimization (Aldridge, 2019), and sentiment analysis (Prabha & Srikanth, 2019). Analysis of major stocks like S&P 500 and Nasdaq-100 over various time periods were conducted (Zhang & Pinsky, 2025) using LSTM model and it was observed that LSTM consistently performed the best, aligning with the model. These findings align with our results,justifying the use of LSTM in stocks. Many such papers (Chen et al., 2023; Nabipour et al., 2020; Selvin et al., 2017) have suggested how deep learning models, particularly LSTM, have significantly improved the prediction of the stock market. Research by Blitz and Huij (2012) and Rompotis (2009) shows that ETFs exhibit strong correlations with their underlying assets. This approach has been further validated in our own work.

We begin the methodology section by presenting a flowchart of the proposed approach (Fig. 1), which provides an overview of the entire process. The pipeline starts with raw data, which undergoes selection and preprocessing to ensure quality and consistency. Next, feature

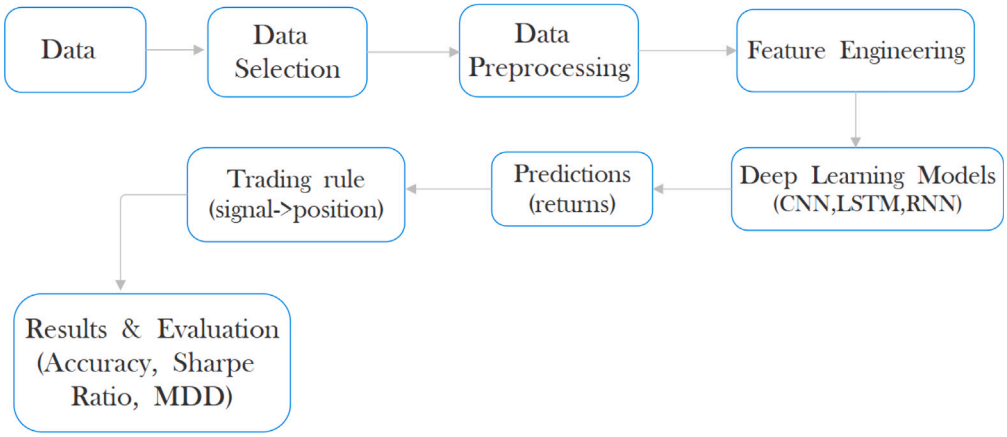


Fig. 1. Flowchart for the modeling process: Data collection → Data preprocessing → Model training → Performance evaluation → Best model selection.

Table 1
Historical ETF data example.

Date	Adj Close	Close	High	Low	Open
03–02–2000	13.3	23.5	23.9	23.3	23.9
04–02–2000	13.3	23.4	23.6	23.2	23.6
07–02–2000	13.1	22.9	23.5	22.9	23.5
08–02–2000	13.1	22.9	23.2	22.7	23
...	...	...	...	...	...

engineering extracts relevant information to enhance model performance. Deep learning models, including CNN, LSTM, and RNN are then trained to generate return predictions. These predictions are translated into trading signals using predefined trading rules and subsequently evaluated using key performance metrics such as accuracy, Sharpe ratio, and maximum drawdown (MDD). This structured workflow enables a systematic and transparent assessment of model effectiveness in predicting stock price movements.

2.1. Data selection and pre-processing

Historical pricing data for ETFs and SPY were collected from Yahoo Finance (yfinance) for the period of January 03, 2000 to December 30, 2024. As part of the data preprocessing technique, daily returns for these ETFs were collected and their true labels T_i were assigned as positive (+) or negative (-) based on the daily return R_i as follows:

$$T_i = \begin{cases} \varepsilon + \varepsilon & \text{if } R_i \geq 0 \\ \varepsilon - \varepsilon & \text{otherwise} \end{cases} \tag{1}$$

where daily return R_i is computed from the closing price P_i and the closing price of the previous day as

$$R_i = \frac{P_i - P_{i-1}}{P_{i-1}} \tag{2}$$

For example, from Table 1, the closing price for a stock is for February 7, 2000 (day i) was $P_i = 22.9$ and the closing price for February 4, 2000 was $P_{i-1} = 23.4$.

Therefore, the daily return R_i is

$$R_i = \frac{22.9 - 23.4}{23.4} = -0.021$$

Since the return is negative ($R_i < 0$), the assigned True label T_i for day i is $T_i = \varepsilon - \varepsilon$. Therefore, with available daily pricing data for any stock S, we can compute the True labels and associate a sequence of these labels to this stock.

We can then ask the question: how good are deep learning models on predicting the label for day i from the sequence of true labels for W preceeding days for a particular stock. Instead of answering this question for a particular stock, we propose to answer this by replacing this stock by the corresponding sector ETF. This approach will be justified by analyzing Correlation and Hamming Distance metrics, as explained in further sections.

As part of our data pre-processing workflow, we examined the class distribution of the target variable to assess whether class imbalance was present. Fig. 2 summarizes yearly model performance using Balanced Accuracy and G-Mean, metrics that offer equitable evaluation across all classes. Throughout the dataset, the class proportions remained relatively stable and reasonably balanced—for example, in 2000–2001, class 0 accounted for about 32%–35% and class 1 for 65%–68% of instances. These consistent ratios minimize the risk of bias in model assessment. The use of Balanced Accuracy (typically between 0.58 and 0.76) and G-Mean (between 0.51 and 0.75) further ensures that performance reflects both classes fairly. This approach ensures meaningful comparison and interpretation of results, even when class proportions differ across time.

2.1.1. Correlations

Correlation is used to measure the strength and direction of linear relationship between two variables. In this paper, correlation is utilized to determine the (linear) relationship strength between individual ETFs and their top holdings. The results would allow us to determine whether the stocks should be replaced with ETFs. This substitution essentially preserves the actual market trend of the stocks while simplifying the model and increasing the computational efficiency of the model. To calculate the correlation, historical data is collected and adjusted closing price of each ETF and its top three holdings over the past 25 years (2000–2024). The correlations are calculated between every ETF and its holding individually, ensuring data consistency by handling missing values.

Fig. 3 illustrates the yearly correlation between ETFs and SPY from 2000 to 2024. Each line highlighted with a different color represents each ETF. This shows how the price movements align closely with SPY over time. The dashed black line at 0.9 indicates a strong correlation threshold. There is a change of values across different years demonstrating the changes in market conditions.

2.1.2. Hamming distance

Hamming Distance is calculated by measuring number of positions in which two sequence of the same length are different. Hamming Distance ratio is the hamming distance normalized by the length of the sequence. In this study, the realized labels of ETFs are compared with their top stock holdings. The hamming distance calculation can be shown using a simple example of two sequences of length 5:

String A:	1	0	1	0	1
String B:	1	0	0	1	1
Hamming:	=	=	≠	≠	=

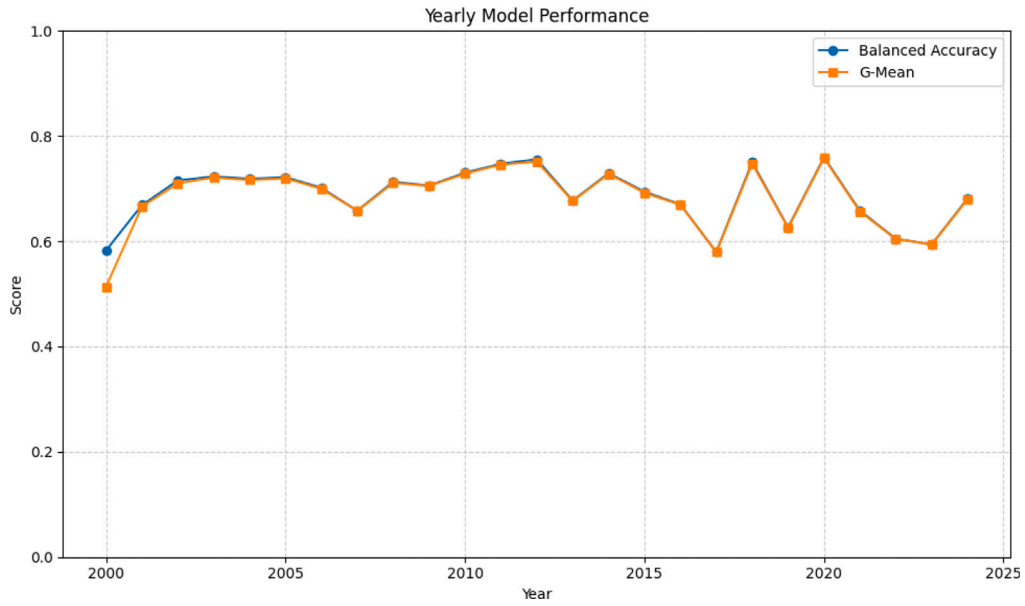


Fig. 2. Yearly model performance (2000–2024), showing Balanced Accuracy (blue line) and G-Mean (orange line) scores. Both metrics evaluate classification performance annually, with higher values indicating better results.

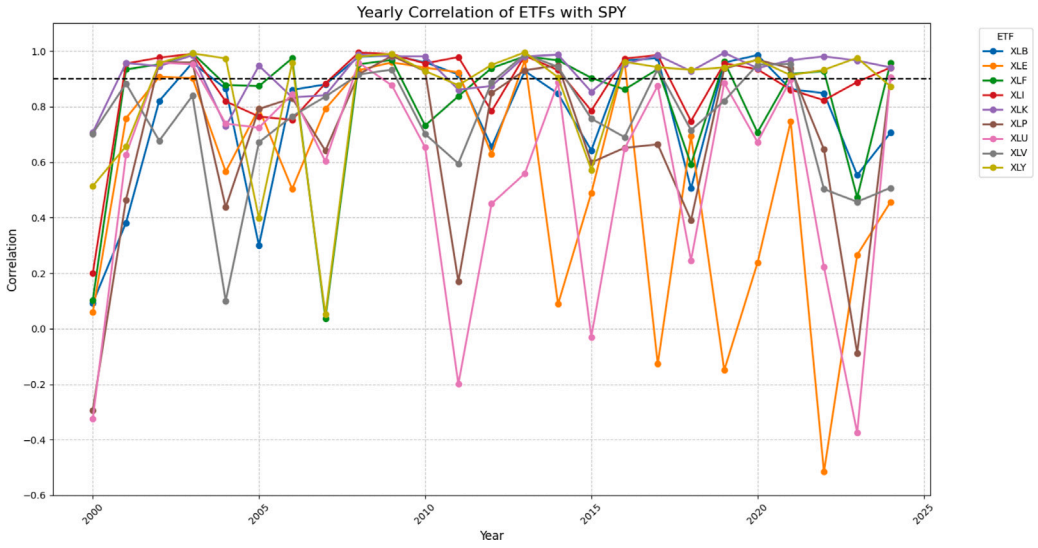


Fig. 3. Correlation between ETFs and SPY over the years.

To calculate the difference in positions, 1 is assigned if there is a mismatch in the strings and 0 if there is a match. As there are 2 differing positions in the example, the hamming distance is 2. The corresponding hamming distance ratio is (2/5) which is 0.4. In the same way, we can compute the hamming distance ratios for ETFs on both input and output sequences after applying the neural network architecture models. This would help us support our argument that holdings are replacable with their corresponding ETFs.

In the next subsections, we provide the detail of our argument.

(A). Input sequence. Initially labels for every stock were computed and ETF by retrieving historical price data from 2000 to 2024 in order to validate our methodology. As already mentioned before, depending on whether the return the following day is positive (+1) or negative (−1) daily returns are calculated and true labels are allocated accordingly. Once the ETFs and their individual holdings have been labeled, it is aligned by date and Hamming distances are calculated to determine how frequently their stock movement direction predictions deviate. ETF can be used as a proxy with certainty if the labels of the ETF and its holdings are quite comparable. Only the top three holdings of each ETF were used in the analysis, as they mostly represented the main constituents of the fund. This approach was chosen because some ETFs did not maintain more than three consistent holdings over the past 25 years. Focusing on the top three holdings – typically the most stable and representative components – allowed for a uniform and meaningful comparison across different ETFs and timeframes. The results are presented in Table 2 and visualized in Fig. 4 for better readability.

The table suggests that the calculated Hamming Distance ratios is very low (averaging 0.25), which shows that the predicted patterns are almost identical to the stocks. Additional tests were conducted for selected ETFs such as XLB, XLK, and SPY using their top five holdings, and the Hamming distance results remained unchanged. This suggests that the number of holdings used per ETF does not significantly impact the analysis and supports the validity of using only the top three holdings. This makes it justifiable to replace the stocks with their ETF

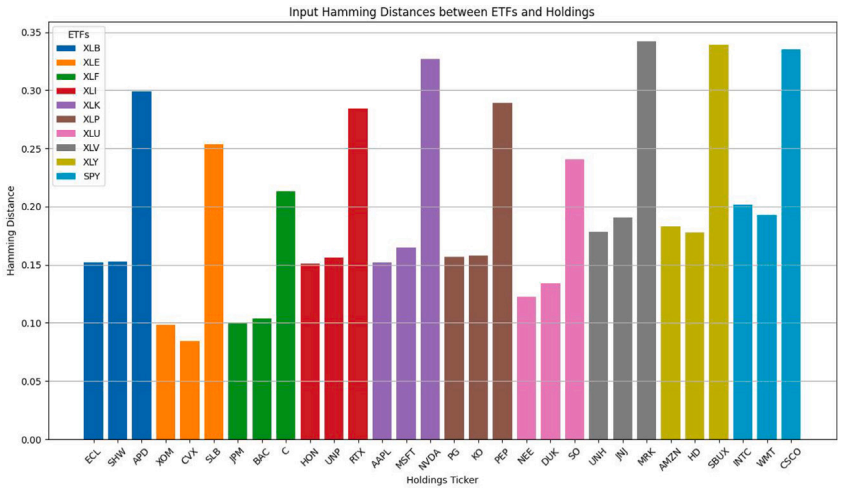


Fig. 4. Hamming Distance Between ETFs and their top 3 Holdings.

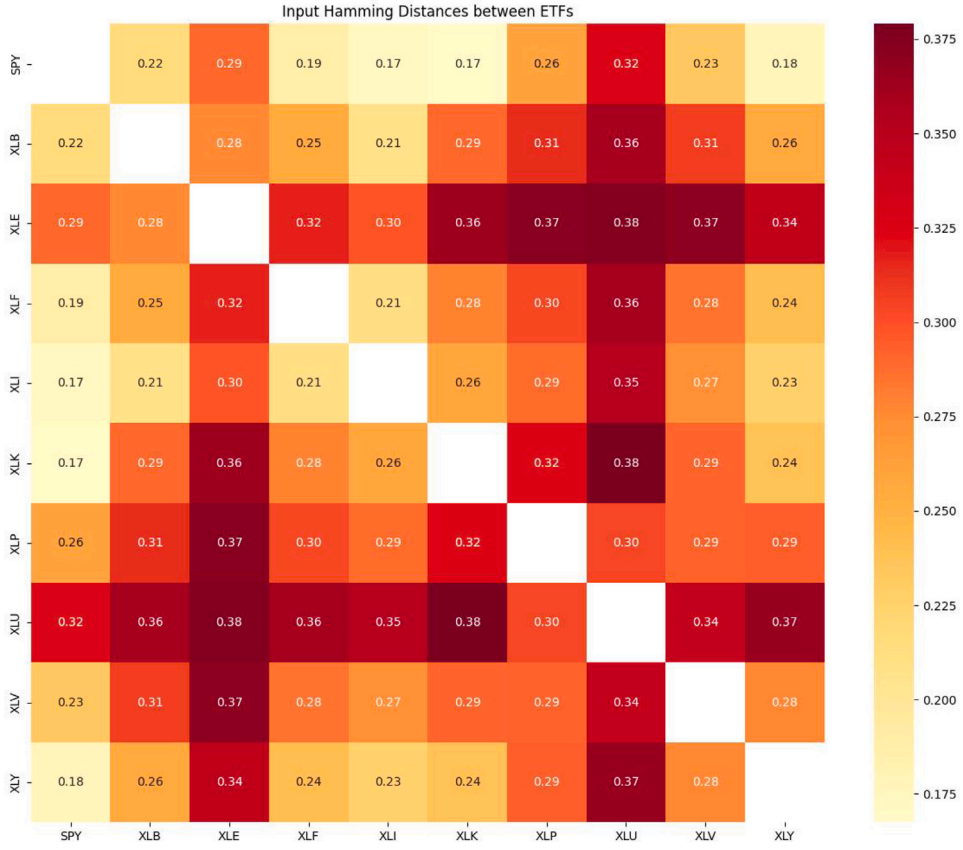
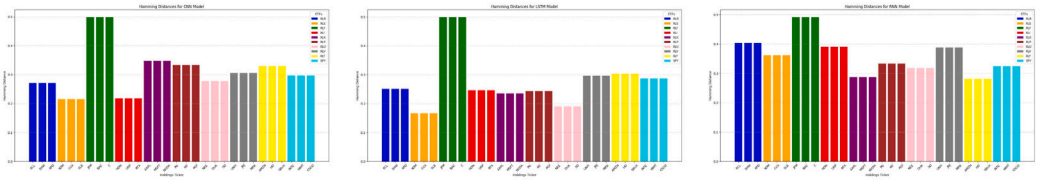


Fig. 5. Input Hamming Distance Between ETFs.

to make the model simpler. By substituting individual stock holdings with their ETFs improves data processing efficiency without sacrificing accuracy in forecasts.

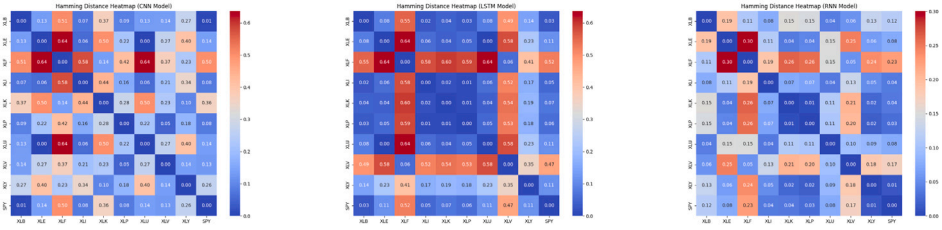
In Fig. 5 the heatmap indicates the Hamming distances between ETF trading signals, measuring their dissimilarity. Higher values (darker colors) suggest more distinct trading patterns, while lower values (lighter colors) imply greater similarity. The trading signals of **XLU** (Utilities) and **XLE** (Energy) are the most distinct since they show the greatest differences from other ETFs. The **SPY** (S&P 500), on the other hand, has comparatively short distances, suggesting that it aligns with several ETFs. Interestingly, **XLE** and **XLP** (Consumer Staples) exhibit a high distance (0.37), indicating notably dissimilar trading patterns.

(B). Output sequence. The process of calculating the output labels is similar to that of the input sequences. However, the predicted daily returns are used to get the labels of ETFs. Even after deep learning techniques are applied, the output sequences produce results that are similar to those of the input sequences. The Fig. 6 displays the output hamming distance ratios between the ETFs and their corresponding holdings using CNN, LSTM and RNN methods, respectively. From the given figures and table attached, it could be depicted that the LSTM visibly had lower hamming distance ratios. This implies that the holdings are highly related, and thus ETFs can proficiently replace the holdings in the output sequence as well.



(a) CNN (b) LSTM (c) RNN

Fig. 6. Output Hamming Distance Between ETFs and their Top 3 Holdings using CNN, LSTM, and RNN.



(a) CNN (b) LSTM (c) RNN

Fig. 7. Output Hamming Heatmap using various models.

Table 2
Input hamming ratios between ETFs and some of its components.

ETF	Ticker	Company name	Input hamming
XLB	ECL	EcoLab	0.15
	SHW	Sherwin Williams	0.15
	APD	Air Prod. and Chemicals	0.30
XLE	XOM	Exxon Mobil	0.10
	CVX	Chevron	0.08
	SLB	Schlumberger	0.25
XLF	JPM	JP Morgan Chase	0.10
	BAC	Bank of America	0.10
	C	Citigroup	0.21
XLI	HON	Honeywell	0.15
	UNP	Union Pacific	0.16
	RTX	Raytheon	0.28
XLK	AAPL	Apple	0.15
	MSFT	Microsoft	0.17
	NVDA	Nvidia	0.33
XLP	PG	Procter & Gamble	0.16
	KO	Coca Cola	0.16
	PEP	Pepsico	0.29
XLU	NEE	NextEra Energy	0.12
	DUK	Duke Energy	0.13
	SO	Southern Co.	0.24
XLV	UNH	United HealthCare	0.18
	JNJ	Johnson & Johnson	0.19
	MRK	Merck	0.34
XLY	AMZN	Amazon	0.18
	HD	Home Depot	0.18
	SBUX	Starbucks	0.34
SPY	INTC	Intc	0.20
	WMT	Walmart	0.19
	CSCO	Cisco	0.34

Table 3
Output hamming ratios for ETF models.

ETF	Model	Tickers	Avg. Hamming
XLB	CNN	ECL, SHW, APD	0.27
	LSTM	ECL, SHW, APD	0.25
	RNN	ECL, SHW, APD	0.40
XLE	CNN	XOM, CVX, SLB	0.22
	LSTM	XOM, CVX, SLB	0.17
	RNN	XOM, CVX, SLB	0.36
XLF	CNN	JPM, BAC, C	0.50
	LSTM	JPM, BAC, C	0.49
	RNN	JPM, BAC, C	0.50
XLI	CNN	HON, UNP, RTX	0.22
	LSTM	HON, UNP, RTX	0.25
	RNN	HON, UNP, RTX	0.39
XLK	CNN	AAPL, MSFT, NVDA	0.35
	LSTM	AAPL, MSFT, NVDA	0.24
	RNN	AAPL, MSFT, NVDA	0.29
XLP	CNN	PG, KO, PEP	0.33
	LSTM	PG, KO, PEP	0.24
	RNN	PG, KO, PEP	0.33
XLU	CNN	NEE, DUK, SO	0.28
	LSTM	NEE, DUK, SO	0.19
	RNN	NEE, DUK, SO	0.32
XLV	CNN	UNH, JNJ, MRK	0.31
	LSTM	UNH, JNJ, MRK	0.30
	RNN	UNH, JNJ, MRK	0.39
XLY	CNN	AMZN, HD, SBUX	0.33
	LSTM	AMZN, HD, SBUX	0.30
	RNN	AMZN, HD, SBUX	0.28
SPY	CNN	INTC, WMT, CSCO	0.30
	LSTM	INTC, WMT, CSCO	0.29
	RNN	INTC, WMT, CSCO	0.32

The highlighted green values in Table 3 indicates lowest average hamming distance value compared to other models for every ETF.

For some ETF pairs in Fig. 7, the LSTM and CNN models in show higher Hamming distances (up to 0.64) as seen in Fig. 7(a) and Fig. 7(b), suggesting more distinct sectoral forecasts. The RNN model, on the other hand in Fig. 7(c), exhibits smaller overall distances (maximum of 0.30), indicating more consistent predictions across ETFs, possibly at the expense of capturing sector-specific complexities.

The distinct market behavior of **XLF** (Financials) and XLE (Energy) is reflected in their persistent higher distances from other ETFs. In contrast, SPY maintains relatively small distances with the majority of ETFs in all models, demonstrating its alignment to general market patterns.

2.2. Feature engineering

The feature engineering process considers the closing prices of ETFs as their primary feature for modeling. The extracted 'Adjusted Close' columns from the downloaded ETF csv files are transformed

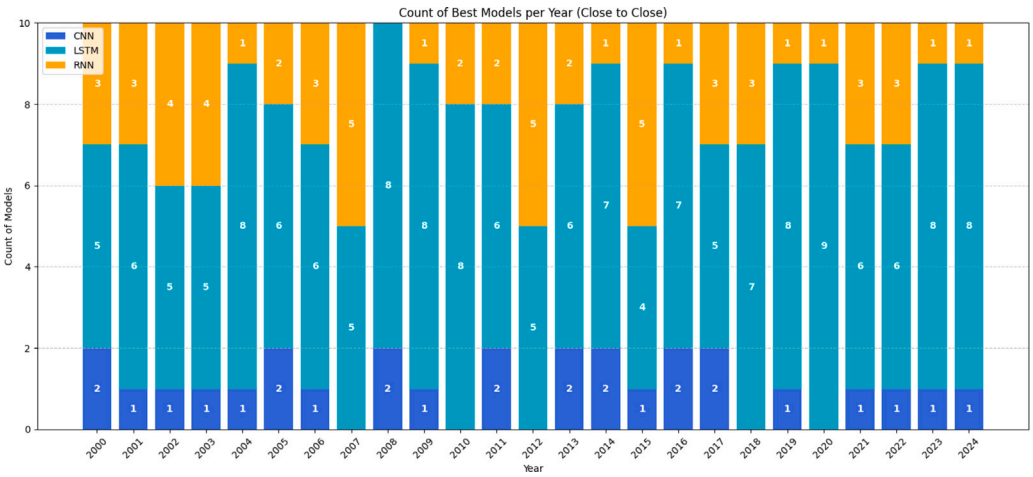


Fig. 8. Yearly count of best-performing models (LSTM, CNN, and RNN) across all ETFs. The graph illustrates which model performed best in each year, highlighting trends in model dominance over time.

into Numpy array for processing. The Min-Max scaling is applied to normalize the data. This increases model performance and reduce outlier impact. Sequential time-series windows based on various lookback periods (10, 30, 60, 180 and 250 days) were created to capture temporal dependencies. Here, each input sequence comprises the past closing prices and assign a binary label to the target variable, indicating whether the next day's closing price rises(1) or falls(0). In financial time-series, lookback 250 is often used as common indicator (Vukovic et al., 2023). Accuracy of the neural architecture models (LSTM,CNN and RNN) were calculated using each lookback value but the performance of the models remained unaffected. Hence, the lookback value of 250 was used to calculate the daily return predicted labels. Finally, the data was effectively transformed for every model architecture. For LSTM and RNN, the input features are reshaped to 3D tensors of shape whereas for CNN the input sequences are structured for 1D convolutional layers. Thus ensuring they can capture local time-series features effectively in price movements.

2.3. Deep learning models

In this paper, three deep learning models is used, Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Convolutional Neural Network (CNN). All of these models are able to handle sequential data well, making them suitable for financial time-series forecasting. First, the dataset is split into training, validation, and test sets, with 50% used for training, 25% for validation, and 25% for testing. We used fixed chronological splits to reflect real-world trading conditions, ensuring models are trained strictly on past data and evaluated on unseen future data. This approach avoids data leakage, simplifies interpretation, and is computationally efficient for large-scale comparisons across different architectures and sectors. The architecture models are then used to evaluate the accuracy of the model and predict their daily return labels. The best-performing model for each year for every ETF is identified and visualized below in Fig. 8. The model that consistently achieved a high Sharpe ratio and low maximum drawdown (MDD) across the majority of sectors was considered the best-performing model. From the graph it can be observed that LSTM consistently outperforms other models across most etfs throughout the years. Hence, proving that LSTM can be used to effectively forecast stock the prices.

All these models use a fixed batch size of 32, Adam optimizer (learning rate = 0.001) and the loss function Binary Cross-entropy.

2.3.1. LSTM

- The LSTM models consists of two LSTM layers with 50 units each. Here, the first layer passes the entire input sequence to the next layer to help capture the long-term dependency.
- Subsequently, a dense layer is used with 25 neurons to extract features. It is followed by an output Dense layer with 1 neuron and sigmoid activation. This helps classify the ETF price movements as positive or negative.

2.3.2. RNN

- Similar to the LSTM layer, the RNN also has two SimpleRNN layers with 50 units each. The first layer passes the entire sequence to the next layer but it may not be able to handle the long-term dependencies as well.
- A dense layer with 25 neurons is added followed by the final Dense layer with 1 neuron and sigmoid activation function to improve feature extraction and predict the price direction accurately.

2.3.3. CNN

- CNN uses the Conv1D layer with 64 filters and a kernel size of 3 to capture changes in stock price movements. This is followed by a MaxPooling1D layer where pool size = 2 to reduce dimensionality and preserve the necessary information.
- The flattened output is passed through a Dense layer consisting of 50 neurons and ReLU activation function to refine the features and then use the final Dense Layer with 1 neuron and sigmoid activation for binary classification of stock price movements.

Table 4 provides a clear overview of the architectures and training hyperparameters for the LSTM, RNN, and CNN models. By presenting these details side-by-side, the table makes it easier to compare the key hyperparameters across different models, supporting a transparent evaluation and reproducibility of the experiments.

2.4. Performance metrics

These metrics are useful to evaluate the model trading strategies and compare the performance based on financial context. Every metric offers an alternative perspective on the risk, return, and overall stability of the model.

Table 4
Summary of model architectures and training hyperparameters.

Model type	Architecture	Activation functions	Training hyperparameters
LSTM	• 2 stacked LSTM layers (50 units each) • Dense layer (25 units) • Dense output layer (1 unit)	• LSTM layers: tanh (default) • Dense(25): linear (no activation) • Output: sigmoid	• Optimizer: Adam (lr=0.001) • Loss: binary_crossentropy • Metrics: accuracy • Batch size: 32 • Epochs: 50 • Validation split: 0.2
RNN	• 2 stacked SimpleRNN layers (50 units each) • Dense layer (25 units) • Dense output layer (1 unit)	• SimpleRNN layers: tanh (default) • Dense(25): linear • Output: sigmoid	• Optimizer: Adam (lr=0.001) • Loss: binary_crossentropy • Metrics: accuracy • Batch size: 32 • Epochs: 50 • Validation split: 0.2
CNN	• Conv1D (64 filters, kernel size 3) • MaxPooling1D (pool size 2) • Flatten layer • Dense layer (50 units) • Dense output layer (1 unit)	• Conv1D: relu • Dense(50): relu • Output: sigmoid	• Optimizer: Adam (lr=0.001) • Loss: binary_crossentropy • Metrics: accuracy • Batch size: 32 • Epochs: 50 • Validation split: 0.2

2.4.1. Volatility

The data is grouped by ETF, model, and year and annualized volatility is calculated. This helps to measure the return fluctuations over the year. The average annualized volatility for each model is calculated and ranked in descending order, thus helping to compare the stability of different prediction models. From the results in Fig. 13 and Table 10, the annualized volatility of strategy returns of the ETFs throughout the years is displayed. LSTM had the lowest average annualized volatility over the years, followed by CNN and RNN.

Although the average return annualized volatility for the models is very similar, we may consider an alternative way to look at this in terms of the number of switches (predicted label changes). Fig. 9 displays the average annual number of switches for each model individually across ETFs. Frequent model switches lead to increased trading activity, which raises transaction costs through commissions, slippage, and bid–ask spreads, potentially eroding profits. In our analysis, the reported performance metrics are gross returns and do not account for transaction costs. Typical costs for ETF trading are in the range of 0.05–0.1% per trade (Quantified Strategies, 2025). The total number of trades executed by each model across all ETFs over 25 years shows that LSTM averages 24 trades per year, whereas CNN and RNN execute approximately 52 and 54 trades annually. Assuming a conservative 0.1% transaction cost per trade, the LSTM strategy incurs substantially lower cumulative costs than CNN or RNN, reducing the impact on net returns, especially in low-margin sectors. This lower trading frequency contributes to reduced annualized volatility, lower trading costs, and improved net performance for the LSTM-based strategy. Therefore, even after accounting for transaction costs, the more stable LSTM model is likely to provide higher net returns and better risk-adjusted performance compared to CNN or RNN, demonstrating the advantage of fewer model switches in practical trading applications.

2.4.2. Annual returns

The total profitibility of an investment can be determined by its annual return. Initially, for each year, the average daily predicted return for each ETF is calculated. To get an accurate estimation, the daily average return is compounded over 252 days, considering that the financial markets typically have 252 trading days over a year. The calculated values for the model's annual returns are attached in the Appendix below in Table 10.

2.4.3. Sharpe ratio

By comparing the annual return with the annualized volatility of the investment, the Sharpe Ratio can help to evaluate the risk-adjusted returns. To calculate, the annualized volatility is extracted. It can be done by creating a product of the daily standard deviation and $\sqrt{252}$. Subsequently, the annual return calculated in the previous step can be divided with this annualized volatility, as in Eq. (3) to calculate the Sharpe Ratio. A higher Sharpe Ratio indicates a better risk-adjusted return. The values can be observed in Table 11.

$$\text{Sharpe Ratio} = \frac{\text{Annual_ Return}}{\text{Annualized_Volatility}} \tag{3}$$

2.4.4. Maximum DrawDown (MDD)

The Maximum Drawdown (MDD) measures the worst peak-to-trough decline in the value of the stocks over time. This metric helps quantify the downside risk by identifying the most significant percentage drop from a peak before the investment recovers. To measure the MDD value, **empyrical.max_drawdown()** function was used to determine and compare the lowest and highest point relatively. A more negative MDD means the investment experienced a larger drop as displayed in Table 11. Lower MDD values are preferred as they indicate lower risk. In most sectors (XLF, XLI, XLK, XLP, XLV and SPY), LSTM-based strategy had lower drawdowns and for most sectors (XLE, XLI, XLK, XLP, XLY and SPY), it had a higher Sharpe's ratio.

Summary of Performance Metrics: These performance metrics help compare and evaluate the performance of the models and guide the path to have better risk assessment and higher profitability. The above results indicate that LSTM perform better than the other two deep learning architectures. To further validate the results, we computed 95% confidence intervals for Annual Returns, Sharpe Ratios, and Maximum Drawdown using the Student's t-distribution to quantify the uncertainty in performance estimates as shown in Table 12. These intervals provide a statistical range indicating where the true metric values are likely to fall, enabling us to evaluate the significance of

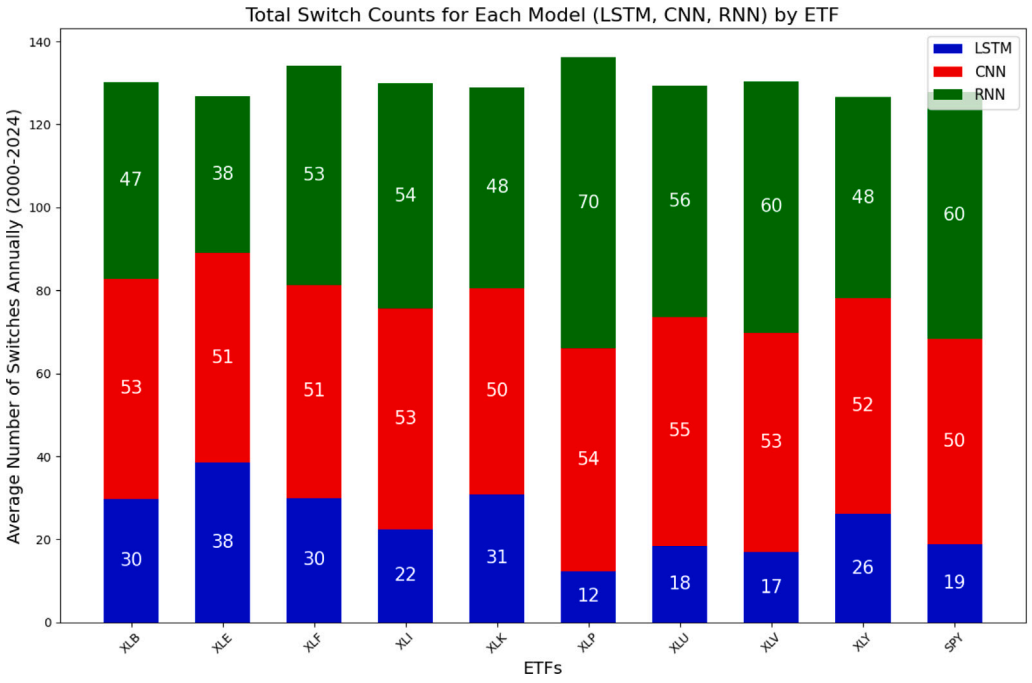


Fig. 9. Average annual number of trades executed per ETF for each model, illustrating differences in trading frequency for LSTM, CNN and RNN.

performance differences between models. The LSTM model consistently showed superior performance with confidence intervals that did not overlap with others, confirming its statistically significant advantage.

3. Results and analysis

This section highlights the financial performance of neural architecture models (LSTM, CNN and RNN) on ETFs. It analyzes the performance of the models and predict their returns. The individual results of the models are compared to Buy and Hold strategy. The main objective of this study intends to evaluate the quality of these deep learning techniques and their investment returns over time. An initial investment of $100 were made at the start of year 2000, to each model along with the Buy and Hold strategy. The daily returns for every ETF were predicted using the models and were used to replicate the portfolio growth over the years. Additionally, true historical ETF price data was utilized to determine the performance of ETFs using Buy and Hold where the stock is simply purchased at the start of the year and held without making active trading decisions. The amount generated is reinvested at the end of the year to the consecutive year for analysis. The ultimate goal of this project is to determine whether the trading algorithms like LSTM,CNN and RNN can yield a higher return on investment than the conventional passive investing strategy. The results depicted that in 80% of our ETFs, **LSTM** model performed the best giving higher investment outcomes. The best performance can be observed in XLK where the final amount generated is $1044 which is 33.14% better than RNN,1672.88% better than CNN and 103.91% better than Buy and Hold strategy, making it significantly better than other models. The generated amount are visualized in Fig. 10. The other model performances has been attached in Fig. 14 for better reference. If effective, these models could be applied to enhance investment decision-making and optimize portfolio management.

The performance of traditional models, including Random Forest and Logistic Regression, was evaluated on the same ETFs for comparison. Key metrics such as Sharpe Ratio, Maximum Drawdown (MDD), and Annual Returns are summarized in Table 5. When compared with the results from deep learning models shown in Table 11, it is evident that LSTM consistently achieves better predictive performance, highlighting the advantages of deep learning approaches over traditional methods in capturing ETF stock movements.

We have also used ensemble models on these traditional methods to compare the performance measures. On applying a Majority Voting Ensemble to traditional models such as Random Forest and Logistic Regression provides more stable and consistent performance across ETFs, providing meaningful benefits for practical trading strategies. From Table 5, we could observe that while the ensemble does not always achieve the highest annual returns, it generally reduces extreme drawdowns and smooths risk, resulting in more balanced risk-adjusted outcomes. Overall, the ensemble proves useful as a stabilizing mechanism, enhancing consistency and robustness of predictions, even if peak returns are sometimes lower than the best-performing individual model.

Furthermore, ensemble methods were also used to evaluate the robustness of our models and to determine whether combining multiple architectures could enhance predictive stability. Ensembles aggregate predictions from multiple deep learning models – LSTM, CNN, and RNN – leveraging their complementary strengths and mitigating individual weaknesses. Two-model hybrids (e.g., LSTM+CNN, LSTM+RNN, CNN+RNN) and the all-three ensemble (LSTM+CNN+RNN) were tested to assess their impact on performance.

Analysis of Table 6 shows that LSTM consistently delivers the strongest performance across most ETFs, while CNN occasionally surpasses it in some cases, and RNN generally performs less effectively. Two-model hybrid models provide more stable results, reducing variability in predictions, and the all-three ensemble offers intermediate but highly consistent performance by smoothing extreme outcomes.

Overall, LSTM remains the most effective single model, while ensemble combinations primarily improve the stability and reliability of predictions across ETFs.

4. Discussion

From the previous results, it was observed that LSTM had achieved comparatively better performance than other two models, CNN and RNN. Table 7 presents a summary of comparisons of different metrics for the three deep learning models and all sectors.

LSTM-based models outperformed all sectors. It significantly outperformed buy-and-hold in XLK (technology) and XLY (consumer discretionary). For other sectors, the out-performance was not much significant.

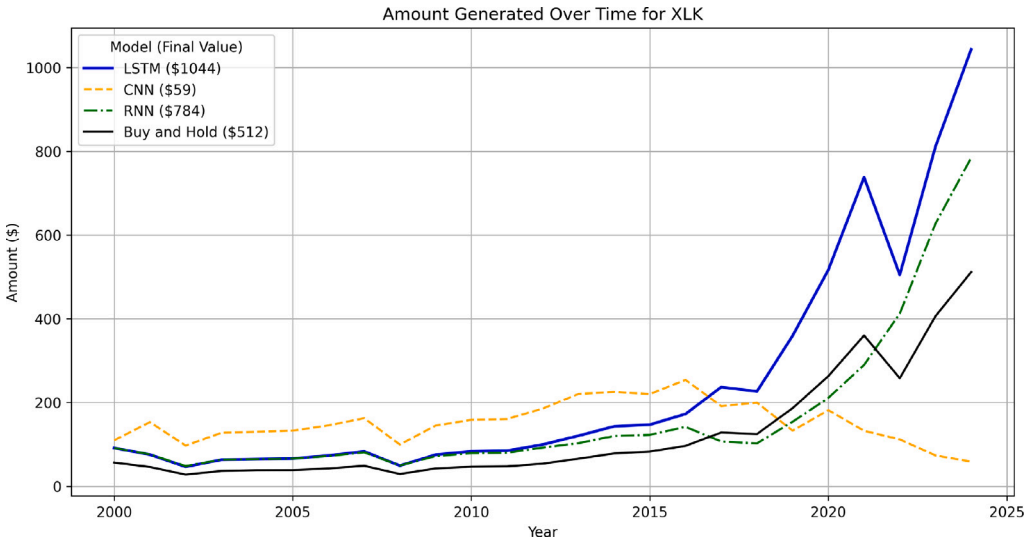


Fig. 10. XLK model Performance using various Deep Learning and Buy and Hold Methods.

Table 5
Performance metrics such as Annual Return(AR), Sharpe Ratio(SR), and Max Drawdown(MDD) for each ETF comparing Random Forest, Logistic Regression, and Majority Voting Ensemble models are shown.

ETF	Random forest			Logistic regression			Majority voting ensemble		
	AR	SR	MDD	AR	SR	MDD	AR	SR	MDD
XLB	0.068	0.485	−26.3	0.064	0.380	−37.6	0.042	0.323	−29.8
XLE	0.099	0.463	−52.4	0.071	0.372	−61.3	0.079	0.403	−52.5
XLF	0.026	0.240	−22.2	0.094	0.474	−43.3	0.016	0.178	−22.2
XLI	0.003	0.095	−33.8	0.101	0.532	−42.8	0.043	0.358	−20.9
XLK	0.050	0.808	−3.7	0.207	0.814	−34.0	0.029	0.559	−4.9
XLP	−0.001	0.034	−24.2	0.045	0.343	−24.9	−0.006	−0.015	−24.5
XLU	0.038	0.356	−13.5	0.018	0.195	−40.2	0.060	0.501	−13.5
XLV	0.017	0.257	−9.5	0.062	0.417	−28.8	0.023	0.334	−9.4
XLY	0.023	0.251	−30.0	0.125	0.584	−40.3	0.018	0.206	−31.8
SPY	0.038	0.427	−10.8	0.128	0.680	−34.1	0.036	0.398	−15.5

Table 6
Comparison of prediction accuracy for individual deep learning models (LSTM, CNN, RNN) versus hybrid models and ensemble combinations across ETFs. The highest accuracy per ETF is highlighted in green.

ETF	LSTM	CNN	RNN	LSTM+CNN	LSTM+RNN	CNN+RNN	All_Three
XLB	0.528	0.522	0.507	0.525	0.517	0.515	0.519
XLE	0.521	0.519	0.507	0.520	0.514	0.513	0.516
XLF	0.472	0.474	0.486	0.473	0.479	0.480	0.477
XLI	0.532	0.532	0.514	0.532	0.523	0.523	0.526
XLK	0.548	0.484	0.521	0.516	0.534	0.502	0.517
XLP	0.532	0.514	0.513	0.523	0.522	0.514	0.520
XLU	0.533	0.518	0.505	0.526	0.519	0.512	0.519
XLV	0.476	0.503	0.493	0.489	0.484	0.498	0.491
XLY	0.523	0.495	0.517	0.509	0.520	0.506	0.512
SPY	0.555	0.547	0.532	0.551	0.543	0.539	0.544

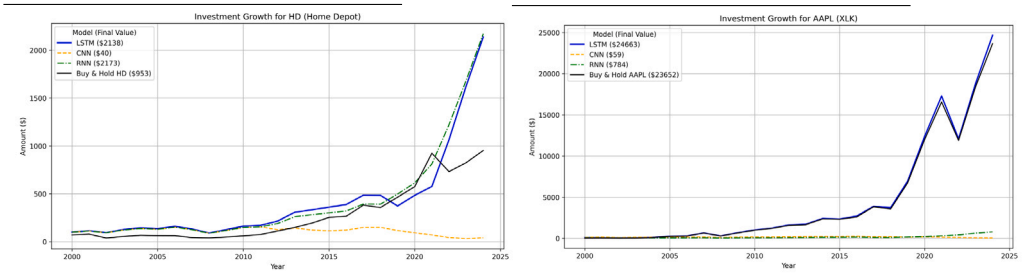
LSTM was found to performed particularly well in high-growth sectors such as Technology (**XLK**) and Consumer Discretionary (**XLY**), with **XLY** achieving 2,131% and **XLK** 1,044% growth, far surpassing other ETFs. Moreover, the Sharpe Ratios for **XLK** (0.94) and **XLY** (0.9) indicate that the model not only captured high returns but also did so with efficient risk-adjusted performance. This suggests that the model effectively identified movements and pattern opportunities in these sectors.

On the other hand, the model struggled with other sectors such as Financials (**XLF**) and Energy (**XLE**), both of which had high annualized volatility (23.9% and 26.4%, respectively) and significantly lower total growth compared to top-performing sectors. The negative Sharpe Ratio of **XLF** (−0.2) highlights the model's inability to generate returns that justified the risk taken in this sector. Additionally, **XLF** and **XLE** experienced the deepest drawdowns (−24.3% and −23.5%), indicating increased vulnerability to market downturns. These results suggest that financial and energy stocks may require additional macroeconomic indicators or alternative modeling techniques to improve predictive accuracy. Although XLE and XLF show lower correlations with constituents, we included them for completeness, as they are standard sector benchmarks. Moreover, both are very important and relevant sectors and removing them would preclude cross-sector comparison. Although XLE has declined has decreased its weights from x to y over the 25 years, Financials still play a big role in the sectors. Furthermore, while **SPY** performed well overall, the LSTM model delivered superior risk-adjusted returns in certain sectors, reinforcing the idea that sector-specific strategies can outperform broad market investments.

Table 7
Performance metrics for Various ETFs using CNN, LSTM, and RNN models.

Metric	Model	XLB	XLE	XLF	XLI	XLK	XLP	XLU	XLV	XLY	SPY
Total Growth	CNN	513	218	80	419	59	157	903	219	40	354
	LSTM	531	519	391	718	1,044	643	688	630	2,131	597
	RNN	106	13	24	43	784	186	59	210	2,173	196
	B&H	514	510	378	690	512	632	678	621	876	570
Annual Return	CNN	10.4	10.7	5.9	10.1	3.7	3.6	12.8	5.8	0.8	8.7
	LSTM	11.7	11.1	7.9	10.2	15.6	5.9	6.5	1.5	16.7	9.8
	RNN	5.1	−2.5	2.7	0.5	15.1	4.2	0.6	5.8	18.1	6.3
Volatility	CNN	20.9	26.4	23.9	20.4	22.7	13.4	18.2	16.9	19.8	16.7
	LSTM	20.9	26.4	23.9	20.4	22.7	13.4	18.2	16.9	19.8	16.7
	RNN	20.9	26.4	23.9	20.4	22.7	13.4	18.2	16.9	19.8	16.7
Sharpe Ratio	CNN	0.64	0.43	−0.09	0.75	0.13	0.35	0.75	0.42	0.14	0.80
	LSTM	0.6	0.5	−0.2	0.76	0.94	0.6	0.57	0.08	0.9	0.88
	RNN	0.33	−0.08	−0.31	0.09	0.78	0.4	0.12	0.45	1.04	0.61
Max DrawDown	CNN	−19.7	−23.7	−22.4	−18.5	−19.9	−12.1	−14.6	−15.6	−21.6	−16.0
	LSTM	−20.1	−23.5	−24.3	−21.1	−22.8	−12.9	−16.5	−17.2	−17.8	−16.8
	RNN	−21.2	−26.5	−22.8	−18.5	−20.0	−12.0	−17.7	−16.0	−17.6	−16.7

Table 8
Performance of XLY and XLK's actual holdings, evaluated using Deep Learning models, based on the constituent stocks HD and AAPL.



(a) HD (XLY- Consumer Discretionary) (b) AAPL (XLK- Technology)

By contrast, RNN provided similar performance for XLY but much lower outperformance for XLK. Finally, CNN significantly underperformed all sectors except for utilities (XLU) as summarized in Table 7. It could be observed that in general LSTMs outperform in high-growth, trend-driven sectors like Technology(XLK) and Consumer Discretionary(XLY) by capturing long-term dependencies, but struggle in volatile, shock-prone sectors like Energy(XLE) and Financials(XLF) where noise and regime shifts limit their effectiveness; tailored architectures may be needed for these complex dynamics.

The performance of the predictive model was evaluated using the actual holdings of selected ETF tickers. As shown in Table 8, the total returns generated using the actual holdings (HD and AAPL) closely mirrored the predictions made using the corresponding ETF tickers (**XLY** and **XLK**) with the deep learning models. This outcome supports our hypothesis that an ETF effectively behaves as a centroid of its underlying holdings and can serve as a reliable proxy for predictive modeling.

The accuracy comparison across ETFs and lookback windows (10, 30, 60, 180 and 250 days) in Fig. 11 shows only marginal differences overall, indicating limited impact of lookback length on model performance. This stability suggests that predictive signals are concentrated in the recent past, reflecting the short-memory nature of most financial sectors and the models' effective capture of essential short-term dependencies. We believe that this does not imply direct insensitivity to temporal context; rather, performance saturates once these key short-term patterns are learned. Notably, XLK (Technology) achieves its best accuracy with a 250-day lookback, likely due to its trend-driven dynamics and slower-changing fundamentals benefiting from longer historical context. Thus, while most sectors require only short-term data, sectors with persistent momentum or structural trends may gain from extended lookback periods.

To further assess robustness, we also conducted sensitivity analysis on additional hyperparameters, namely the number of training epochs and batch sizes, for three representative ETFs(XLB,XLE,XLF) as shown in Table 13. Across variations (epochs = 50, 100, 150; batch sizes = 32, 64, 128), the models achieved nearly identical predictive accuracies. Taken together with the lookback experiments, these results suggest that model performance is stable across a range of hyperparameter settings. This consistent results provide evidence that our findings are robust to reasonable hyperparameter changes, and the overall conclusions of the trading strategy evaluation remain valid.

To evaluate model performance in terms of predictive accuracy and assess potential overfitting, we computed the yearly average train and test accuracies for each ETF. Across ETFs, mean overfitting ranges from approximately -3% to +1.2%, with most ETFs showing minimal overfitting, while a few (e.g., XLK, XLE) exhibit slightly positive overfitting up to 2.7% in some years. A 3D bar plot as displayed in Fig. 12 was generated, with the x-axis representing years, the y-axis representing ETFs, and the z-axis representing accuracy. Separate bars were plotted for training and test sets, allowing a direct visual comparison of performance across time and sectors. The resulting plot indicated that test accuracy consistently matched or exceeded training accuracy for all ETFs, with no significant gaps, suggesting that the models were not overfitting and were generalizing well across the dataset.

5. Contributions and limitations

5.1. Novelty

The novelty of the paper lies in the idea that the individual stock holdings can be replaced with ETF stock data. This simplifies the input and makes the model more efficient. The validity of using the

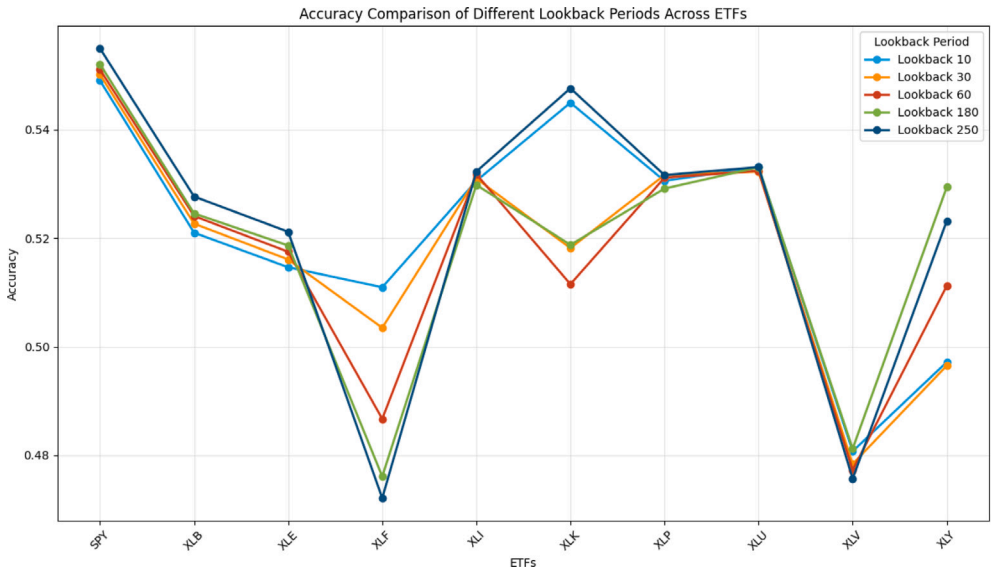


Fig. 11. Comparison of accuracy using various lookback values across ETFs.

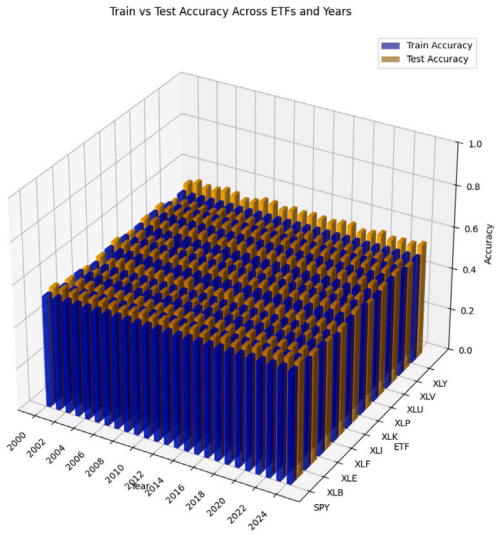


Fig. 12. Comparison of yearly train and test accuracies for each ETF, showing minimal overfitting.

ETFs is upheld by the low hamming distances and high correlation between the stock holdings and ETFs. In line with both industry practice and academic convention, we use an average correlation threshold of 0.995 and Hamming distance 0.25 as indicative of near-perfect substitution between sector ETFs and their representative stocks. Prior research on ETF substitutability, arbitrage, and risk management recognizes correlations over 0.99 as denoting practical equivalence for portfolio purposes, and variable Hamming distance cutoffs are typically chosen to reflect strong overlap in signal or classification agreement (Dambra et al., 2025; Gudmundarson & Peters, 2024). Our selected thresholds are data-driven, reflecting a clear separation between high-correlation/high-agreement ETFs and outlier cases, and are consistent with both financial econometrics literature and common portfolio management practices. Furthermore, unlike previous studies that predominantly focused on broader industry-level indices, our analysis leverages individual sector ETFs, enabling the models to exploit finer-grained patterns for trading strategies. ETFs aggregate underlying stock behavior, lowering volatility exposure and enabling models to capture broader market or sector trends more effectively. This sector-level focus allows for more precise forecasting of price movements, while still supporting general index strategies, demonstrating that deep learning models – particularly LSTM – can enhance both prediction accuracy and investment performance when applied to sector ETFs rather than individual stocks.

5.2. Black swan events

Since our study spans over 25 years (2000–2024), we were able to assess the impact of major "black swan" events on model performance during specific years—namely, the **Dot-com bubble (2001)**, the **Global financial crisis (2009)**, and the **COVID-19 pandemic (2019)**. We focused on the sectors most affected in each case: XLK (Technology) in 2001, XLF (Financials) in 2009, and XLE (Energy) in 2019.

In our approach, the best-performing model for each ETF and year was determined by comparing the final compounded investment amount generated by each of the LSTM, CNN, and RNN models. The model with the highest return in a given year was marked as the best, and this comparison was conducted annually. Table 9 visualizes how often each model outperformed the others across time for these ETFs.

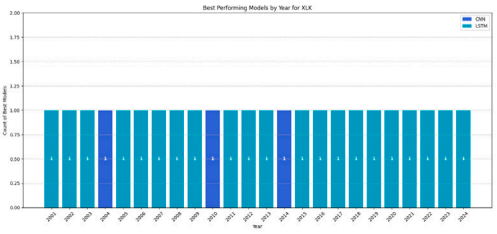
Interestingly, in 2001, the LSTM model performed the best for XLK, although this may be due to the limited availability of training data before the dot-com crash, which could have inadvertently favored LSTM's generalization over the other models. In contrast, during the 2009 financial crisis, LSTM failed to capture the sharp decline in the financial sector (XLF), leading to underperformance. Similarly, in 2019, amidst extreme volatility in the energy sector (XLE), the RNN model outperformed others, suggesting greater adaptability to abrupt market shifts.

These observations indicate that while LSTM models generally excel in capturing long-term dependencies, they may struggle during periods of high uncertainty and structural breaks, such as black swan events. In such scenarios, simpler or less memory-intensive models like CNNs or RNNs may offer more robust performance due to their ability to adapt more quickly to rapid changes in market dynamics.

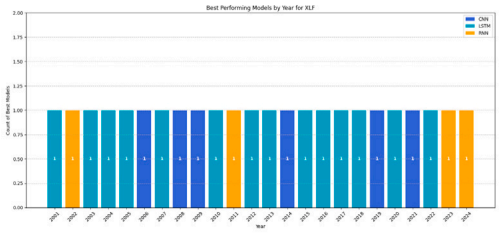
5.3. Comparison with previous work

This paper focuses on long-term (2000–2024) model evaluation compared to models in the existing literature discussed in Section 1 that mostly focuses on short term dependency. This long-term analysis strengthens the validity of the results, ensuring better performance for

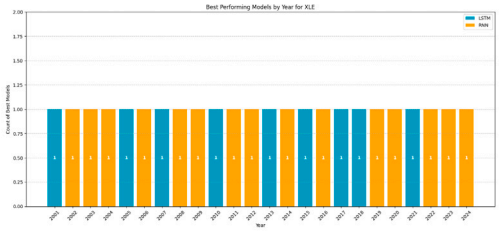
Table 9
Black Swan Events throughout the years (2000–2024) and the model that performed the best during individual years.



(a) Dot-com bubble 2001 (XLK - Technology)



(b) Global financial crisis 2009 (XLF - Financial)



(c) Covid-19 pandemic 2020 (XLE - Energy)

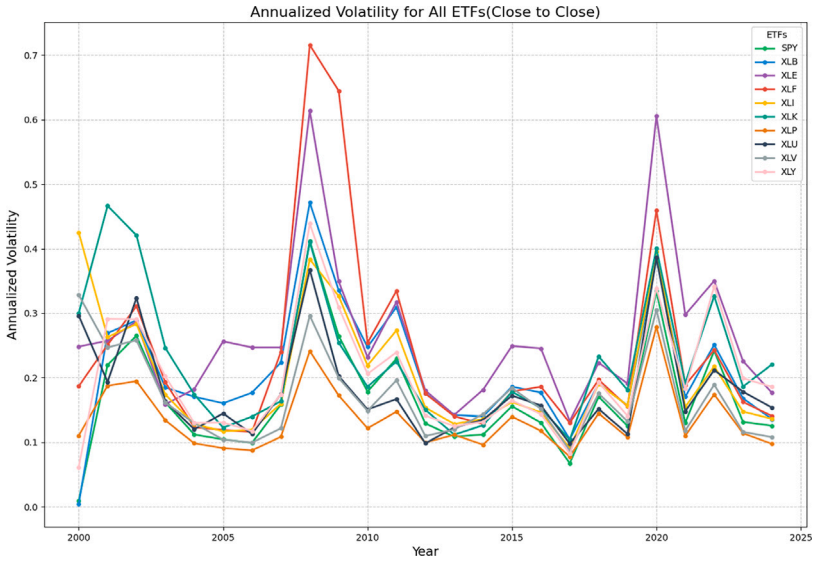


Fig. 13. Annualized Volatility for all ETFs (Close to Close)

models under various economic conditions, including financial crises. Moreover, many studies evaluate their models based on prediction accuracy alone. By contrast, our model incorporates additional financial performance metrics such as Sharpe ratio, annualized volatility, and maximum drawdown. These additional measures provide a more realistic assessment of the models' applicability to real-world trading and investment strategies. We believe that our proposed research provides a useful and actionable approach by taking into account both predictive performance and financial viability.

5.4. Limitation

Despite the promising results, the study has some limitations. The analysis is dependent on historical data which are influenced by market volatility. While LSTM performs well on the data and is good at recognizing patterns, it might not react well to sudden changes in the market. We hope to investigate in more detail in our future work.

6. Conclusion

The paper demonstrates that LSTM model consistently outperformed CNN, RNN and even traditional strategies like Buy and Hold while significantly improving the prediction of stock price movements for ETFs and the S&P 500 data. Overall, the study corroborates the potential of LSTM in stock price prediction by providing a data-driven strategy that could improve portfolio management techniques. Based on the results presented in the Discussion section, we conclude that, apart from specific sectors such as Technology and Consumer Discretionary, the application of deep learning models for stock prediction is generally not recommended, as the performance tends to be inconsistent or

suboptimal across other sectors. **XLK** (Technology) and **XLY** (Consumer Discretionary) tend to perform well in deep learning models because they exhibit clearer trends and momentum, which aligns well with how LSTMs learn patterns over time. These sectors are also more influenced by behavioral and seasonal factors – such as retail trading, earnings cycles, and holiday spending – which create predictable signals. In contrast, sectors like energy (**XLE**) are often too noisy, financials (**XLF**) are heavily impacted by macroeconomic events, and utilities or staples (**XLU**, **XLP**) are too stable, making them less suitable for dynamic prediction models.

This ETF-centric approach can be applied to non-US markets and non-equity assets like cryptocurrencies as well. However, adaptations are needed to account for different market structures, liquidity, and volatility. Emerging markets may exhibit higher noise and lower data availability, requiring tailored preprocessing and modeling. Cryptocurrencies' unique behaviors and 24/7 trading demand models that handle rapid regime changes and higher volatility. Overall, this approach is flexible but must be customized for each asset class and market environment. We hope to explore these areas in near future as well.

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Declaration of competing interest

We have no financial interests to disclose.

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Appendix A. Annual returns and annualized volatility

See Table 10.

Appendix B. Sharpe ratios and MDD

See Table 11.

Appendix C. Annual returns, sharpe ratios and MDD confidence interval

See Table 12.

Appendix D. Comparison of models

See Figs. 13 and 14.

Appendix E. Robustness of model accuracy to epochs and batch sizes

See Table 13.

Data availability

All the relevant data, Python code for analysis, detailed annual tables and graphs are available via: https://github.com/triparnakundu/stock_prediction.

Table 10
Annual returns and annualized volatility metrics for ETF models.

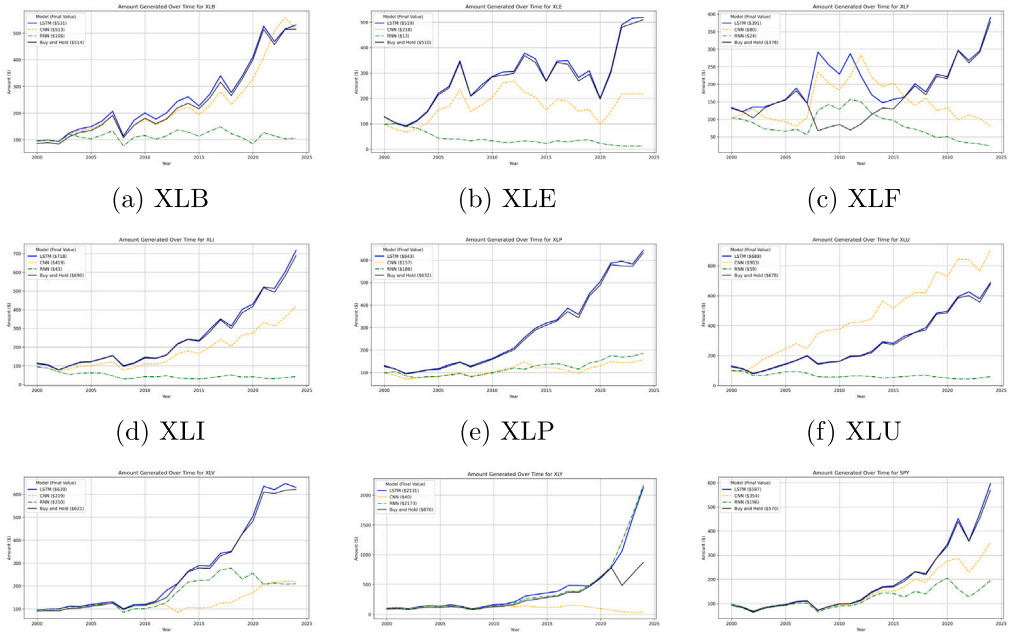
ETF	Model	annualized volatility (%)			Annual Returns (%)		
		Avg	Med	Std	Avg	Med	Std
XLB	CNN	20.9	17.7	9.4	10.4	13.6	18.3
	LSTM	20.9	17.7	9.4	11.7	12.9	18.3
	RNN	20.9	17.7	9.4	5.1	0.6	20.6
XLE	CNN	26.4	24.6	11.6	10.7	4.9	26.4
	LSTM	26.4	24.6	11.6	11.1	7.0	25.4
	RNN	26.4	24.8	11.7	−2.5	−5.5	19.1
XLF	CNN	23.9	18.7	15.1	5.9	−5.8	42.0
	LSTM	23.9	18.7	15.2	7.9	−6.9	42.2
	RNN	23.9	18.7	15.2	2.7	−7.2	42.8
XLI	CNN	20.4	16.0	9.3	10.1	13.2	16.5
	LSTM	20.4	16.0	9.3	10.2	13.2	16.5
	RNN	20.4	15.9	9.3	0.5	0.8	18.1
XLK	CNN	22.7	18.6	10.2	3.7	3.8	26.4
	LSTM	22.7	18.7	10.2	15.6	15.0	24.8
	RNN	22.7	18.7	10.2	15.1	12.8	25.5
XLP	CNN	13.4	11.4	4.8	3.6	7.3	11.4
	LSTM	13.4	11.4	4.8	5.9	8.3	10.2
	RNN	13.5	11.4	4.8	4.2	7.2	10.1
XLU	CNN	18.1	15.7	7.7	12.8	10.9	17.5
	LSTM	18.2	15.7	7.8	6.5	7.5	14.2
	RNN	18.2	15.7	7.8	0.6	1.4	13.7
XLV	CNN	16.9	14.9	6.8	5.8	4.8	12.2
	LSTM	16.9	14.9	6.8	1.5	2.1	13.7
	RNN	16.9	14.9	6.7	5.8	4.8	13.7
XLY	CNN	19.9	18.3	8.9	0.8	−1.4	21.3
	LSTM	19.8	18.3	8.9	16.7	17.4	23.4
	RNN	19.8	18.2	8.9	18.1	20.1	19.8
SPY	CNN	16.7	13.1	8.5	8.7	12.5	15.9
	LSTM	16.7	13.1	8.5	9.8	12.9	16.4
	RNN	16.7	13.1	8.5	6.3	7.3	17.3

Table 11
Sharpe ratio and maximum drawdown for ETF models.

ETF	Model	Sharpe Ratio			Max.Drawdown		
		Avg	Med	Std	Avg	Med	Std
XLB	CNN	0.64	0.71	0.82	−19.7	−16.2	11.5
	LSTM	0.60	0.70	0.82	−20.1	−16.2	11.5
	RNN	0.33	0.08	0.97	−21.2	−16.8	11.7
XLE	CNN	0.43	0.26	0.98	−23.7	−18.6	12.7
	LSTM	0.50	0.28	0.94	−23.5	−18.6	13.2
	RNN	−0.08	−0.20	0.80	−26.5	−26.7	11.1
XLF	CNN	−0.09	−0.34	1.11	−22.4	−19.3	11.3
	LSTM	−0.20	−0.44	1.06	−24.3	−20.0	10.2
	RNN	−0.31	−0.42	0.92	−22.8	−24.7	9.5
XLI	CNN	0.75	0.83	0.95	−18.5	−12.8	12.9
	LSTM	0.76	0.84	0.95	−21.1	−12.8	12.8
	RNN	0.09	0.03	1.09	−18.5	−20.0	12.4
XLK	CNN	0.13	0.23	1.16	−19.9	−18.6	12.5
	LSTM	0.94	0.95	1.14	−22.8	−15.7	14.4
	RNN	0.78	0.94	1.17	−20.0	−16.2	13.8
XLP	CNN	0.35	0.45	0.96	−12.1	−10.2	7.4
	LSTM	0.60	0.73	0.80	−12.9	−9.7	7.1
	RNN	0.40	0.43	0.80	−12.0	−9.9	6.8
XLU	CNN	0.75	0.83	0.90	−14.6	−11.8	7.3
	LSTM	0.57	0.65	0.84	−16.5	−11.9	10.9
	RNN	0.12	0.10	0.83	−17.7	−12.7	10.6
XLV	CNN	0.42	0.28	0.92	−15.6	−13.7	8.2
	LSTM	0.08	0.15	1.07	−17.2	−15.9	8.2
	RNN	0.45	0.28	1.00	−16.0	−15.5	8.9
XLY	CNN	0.14	−0.10	1.17	−21.6	−18.7	11.7
	LSTM	0.90	0.99	1.15	−17.8	−13.4	10.8
	RNN	1.04	1.13	1.02	−17.6	−13.4	10.9
SPY	CNN	0.80	0.77	1.03	−16.0	−10.7	11.4
	LSTM	0.88	0.88	1.07	−16.8	−10.1	11.5
	RNN	0.61	0.58	1.18	−16.7	−13.0	11.3

Table 12
95% confidence intervals for performance metrics, highlighting the best values per ETF — highest Annual Return (Upper CI) and Sharpe Ratio (Upper CI) in green, and lowest Maximum Drawdown (Lower CI) in green.

ETF	Model	Annual Return CI		Sharpe Ratio CI		Max. Drawdown CI	
		Lower	Upper	Lower	Upper	Lower	Upper
XLB	cnn	0.0404	0.1825	0.3032	0.9554	−0.2459	−14.87
	lstm	0.0266	0.1966	0.2564	1.003	−0.2497	−15.25
	rnn	−0.0364	0.1391	−0.0830	0.7432	−0.2613	−16.22
XLE	cnn	−0.0038	0.2209	0.0160	0.8531	−0.2904	−18.29
	lstm	0.0043	0.2206	0.1043	0.9072	−0.2910	−17.98
	rnn	−0.1067	0.0562	−0.4193	0.2663	−0.3116	−21.81
XLF	cnn	−0.0994	0.2599	−0.5612	0.3849	−0.2718	−17.63
	lstm	−0.1197	0.2407	−0.6576	0.2520	−0.2657	−17.99
	rnn	−0.1550	0.2108	−0.7052	0.0786	−0.2882	−17.63
XLI	cnn	0.0313	0.1721	0.3541	1.1658	−0.2397	−13.07
	lstm	0.0332	0.1734	0.3615	1.1698	−0.2354	−12.74
	rnn	−0.0713	0.0825	−0.3762	0.5580	−0.2667	−16.23
XLK	cnn	−0.0752	0.1500	−0.3635	0.6267	−0.2815	−17.61
	lstm	0.0466	0.2582	0.4704	1.4422	−0.2591	−13.78
	rnn	0.0486	0.2657	0.2950	1.2935	−0.2586	−14.21
XLP	cnn	−0.0119	0.0856	−0.0556	0.7675	−0.1609	−9.83
	lstm	0.0161	0.1029	0.2656	0.9469	−0.1513	−9.16
	rnn	−0.0002	0.0855	0.0590	0.7471	−0.1491	−9.16
XLU	cnn	0.0544	0.2037	0.3800	0.9349	−0.1771	−11.53
	lstm	0.0047	0.1258	0.2209	1.1490	−0.2115	−11.53
	rnn	−0.0518	0.0645	−0.2320	0.4778	−0.2216	−11.96
XLV	cnn	0.0066	0.1109	0.0282	0.8171	−0.1903	−12.10
	lstm	−0.0429	0.0739	−0.3731	0.8788	−0.2069	−12.10
	rnn	0.0004	0.1173	0.0235	0.5428	−0.1984	−13.79
XLY	cnn	−0.0821	0.0990	−0.3539	0.6489	−0.2656	−16.72
	lstm	0.0683	0.2682	0.4209	1.4016	−0.2236	−13.20
	rnn	0.0985	0.2671	0.6125	1.4883	−0.2227	−13.01
SPY	cnn	0.0205	0.1554	0.3677	1.2483	−0.2081	−11.22
	lstm	0.0286	0.1685	0.4319	1.3482	−0.2065	−10.91
	rnn	−0.0097	0.1378	0.1150	1.1206	−0.2171	−12.17



(a) XLB (b) XLE (c) XLF

(d) XLI (e) XLP (f) XLU

(g) XLV (h) XLY (i) SPY

Fig. 14. Model Performance Comparison for every ETFs displaying the total growth over the years for each model (LSTM, CNN, RNN) along with their Buy and Hold values.

Table 13
Sensitivity of predictive accuracy to epochs and batch sizes for representative ETFs (XLB, XLE,XLF). Minimal variation indicates robustness to these hyperparameters.

ETF	Epochs	Batch 16	Batch 32	Batch 64
XLB	20	0.5151	0.5189	0.5105
	50	0.5196	0.5176	0.5131
	100	0.5160	0.5144	0.5145
XLE	20	0.5170	0.5151	0.5177
	50	0.5163	0.5170	0.5154
	100	0.5160	0.5097	0.5125
XLF	20	0.4761	0.4880	0.4761
	50	0.4990	0.4761	0.4912
	100	0.4763	0.5014	0.5003

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